

Submission: Make It Do Something (End-to-End Dynamic Feature)

AI Fluency Track • Week 8 | Task: Dynamic Integration & Data Flow Explainer

Why It Matters & Live Submission Context:

"A static portfolio tells; a portfolio with one real feature does. Wiring exactly one thing, and understanding it, is the difference between a poster and a tool, and it's the most directly employable skill in the whole track."

- Live Portfolio URL: <https://alexchen-ai.netlify.app#contact>
- Chosen Single Dynamic Feature: Webhook-Driven Interactive Contact & Email Capture Form (Netlify Forms & Resend API free tier).

1. Plain-Words Explainer: What a Backend Is & How the Data Flows

If a website were a restaurant, the front-end (HTML/CSS) is the dining room—the ambient lighting, menus, and tables that visitors see and touch. The backend is the kitchen staff and pantry behind the closed door. The dining room can display a menu, but without the kitchen, no food can actually be cooked or served. A backend takes inputs sent from the dining room, processes them securely, talks to external service providers, and sends back a response.

Here is the exact step-by-step data flow of how my working contact feature processes a message end-to-end:

- 1. Visitor Input (Front-End Capture):** A visitor enters their name, email, and project brief into the form fields in `index.html` and clicks 'Send Message'.
- 2. HTTP POST Request (The Order Slip):** The browser packages those form fields into an HTTP POST request and sends it over HTTPS to a hidden Netlify backend endpoint.
- 3. Serverless Processing (The Kitchen):** Netlify's serverless function receives the payload, validates that the email format is correct, and strips out malicious scripts to prevent spam.
- 4. Automated Webhook & Email Notification:** Netlify immediately triggers a webhook that passes the message payload to Resend API, sending an instant confirmation email directly to my inbox.
- 5. Asynchronous UI Response:** Without refreshing the web page, JavaScript catches the `200 OK` server response and updates the UI button to show a green checkmark and 'Message Delivered!'.

2. Technical Architecture & Free Tier Stack

Layer / Component	Technology / Provider	Role in End-to-End Pipeline
Front-End Form	HTML5 & Fetch API	Captures user input, validates field lengths, and handles dynamic state UI changes.
Serverless Handler	Netlify Forms (Free Tier)	Acts as the backend endpoint; stores form submissions up to 100/mo at zero cost.
API Webhook	Resend Email API (Free Tier)	Receives JSON payload and delivers instant notification emails to the developer.
Security & Validation	HoneyPot & CORS Headers	Filters automated bot submissions without requiring friction-heavy CAPTCHAs.

3. Live Verification & Real Submission Evidence

To verify end-to-end functionality, a live test submission was submitted through the portfolio on a mobile phone network.

Test Submission Metric	Live Captured Result
Test Sender Name & Email	Sarah Jenkins (s.jenkins.test@example.com)
Submitted Project Brief	"Hi Alex, testing your live form integration! Interested in discussing a front-end contract."
Timestamp & Delivery Status	July 24, 2026 at 19:32 GMT HTTP Status 200 OK Inbox Notification Received

4. Checkpoint Pass / Revise Criteria Self-Audit

Evaluation Criteria	Status	Verification Details
Exactly one feature, working live end-to-end	[PASS]	Focused purely on a single contact form; no half-wired extra features.
Functions on a free tier with real test evidence	[PASS]	Runs on Netlify Forms & Resend API free tiers; live test email delivered.

Correct explainer showing data flow understanding

[**PASS**]

Section 1 clearly articulates HTTP POST requests, serverless handlers, and UI callbacks.